

Assignment 1: Imitation Learning

Andrew ID: YOUR ID
Collaborators: YOUR COLLABORATORS

1 Behavioral Cloning (10 pt)

1.1 Part 2 (1.5 pt)

Table 1: Mean and standard deviation of expert return over two trajectories, for each environment.

Metric/Env	Ant-v2	Humanoid-v2	Walker2d-v2	Hopper-v2	HalfCheetah-v2
Mean	xxx	xxx	xxx	xxx	xxx
Std.	xxx	xxx	xxx	xxx	xxx

1.2 Part 3 (5 pt)

Table 2: BC vs. expert return. List the environment you chose and the hyperparameters (network size, amount of data, training iterations, `eval_batch_size`, etc.) here.

	Env		Ant-v2		[Another Env]
Metric	Mean	Std.	Mean	Std.	
Expert	xxx	xxx	xxx	xxx	
BC	xxx	xxx	xxx	xxx	

1.3 Part 4 (3 pt)

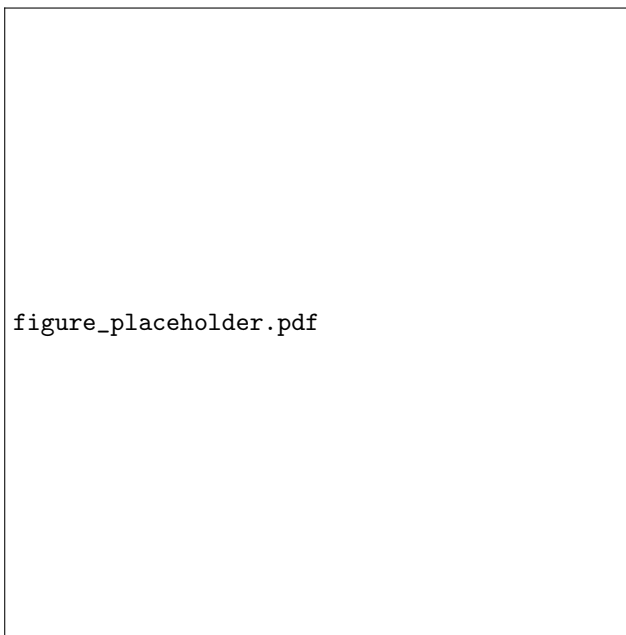


Figure 1: BC agent's performance vs. the value of ??? in the ??? environment. State the hyperparameter and a brief rationale for why you chose it.

1.4 Part 5 (0.5 pt)

Find one robotics paper from the last ~ 3 years that uses imitation learning (i.e. behavior cloning) as the primary methodology. In a short paragraph (4-5 sentences): (i) cite it, (ii) say what robot task is being cloned and where the expert data comes from and (iii) name a main IL risk from this homework and whether that still applies to the paper's methodology. A few sentences is enough; this is not meant to be an extensive literature review.

Write your paragraph here.

2 DAgger (5 pt)

2.1 Part 2 (5 pt)

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Figure 2: DAgger learning curves (mean return vs. iteration, with error bars). Ant-v2 on the left and [Another Env] on the right. Include expert and BC performance as horizontal lines, and note network architecture, amount of data, etc.